NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Assignment 1

Scenario 4: IT Components Problem Reporting System

Group Members:

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INTRODUCTION & PROJECT SCOPE:

Our team has chosen Scenario 4 for this project in order to create a "IT Components Problem Reporting System." The university's current procedure for reporting defective hardware, such as a dead network cable, a malfunctioning projector in a classroom, or a broken PC in a computer lab, is primarily manual and ineffective. Usually, instructors or students must either write down the problem in a physical register or verbally notify the IT department. This results in lost complaints, delayed responses, and no visibility into the repair's progress.

Our suggested system is a digital memo-style platform that is centralized. A simplified web-based system that allows users to quickly submit a digital memo outlining the hardware problem is intended to replace the manual registers. This system will be used by the main stakeholders, the IT staff, to efficiently receive, prioritize, track, and fix hardware issues across the campus.

REQUIREMENT ELICITATION:

We had to collect requirements directly from those who encounter these issues and those who resolve them in order to fully comprehend the needs of the system. To collect our raw data, we employed the following elicitation techniques:

- Interviews with Major Stakeholders (IT Staff):

We had casual talks with the IT support team on campus. Their main complaint was that it is hard to prioritize tasks when you have to deal with complaints from many different sources, such as phone calls, walk-ins, and messages from faculty. They specifically asked for a system that sorts problems by location (like certain computer labs or faculty offices) and type of equipment.

- Observation:

We watched how students are acting in the programming labs right now. A lot of the time, if a student sits at a computer with a broken mouse or a boot problem, they just move to a different one without saying anything. The broken computer stays unused until a lab teacher finally sees it. This observation made it clear that the reporting process needs to be very quick and easy.

- Brainstorming:

Our group had a brainstorming session to come up with user-friendly features because we are students who use campus hardware a lot. We decided that users need a simple dropdown menu to pick the lab and PC number instead of typing out long descriptions.

REQUIREMENT ANALYSIS:

After the elicitation stage, we examined the raw data to distinguish between general user preferences and actual system requirements and to identify the essential features required to address the issues found.

Analysis of the Current Flaws:

- A standardized reporting channel does not exist.
- There isn't a digital log that the IT staff can use to pinpoint recurrent hardware malfunctions in particular labs.

- There is no indication to the person who reports the problem whether it has been acknowledged or fixed.

Proposed system architecture & stakeholder roles:

According to our analysis, the system needs to serve as an effective conduit between the IT department and the complainants. Three main actors for the system have been identified:

- IT Staff:

To view all incoming memos, filter them by location or urgency, and update the ticket status (e.g., "Pending," "In Progress," "Resolved"), IT staff (Primary/Major Stakeholders) need a central dashboard.

- Users (Students & Faculty):

They need an easy-to-use interface that allows them to quickly log a memo with the location, equipment ID, and problem description. They also need to be able to monitor the status of their memo.

- System Admin:

Needs backend access to view overall resolution reports and manage IT staff accounts.

DOCUMENTATION STRATEGY:

The insights from our elicitation and analysis have been converted into formal software requirements to guarantee that the development phase is built on a strong foundation. These requirements are strictly divided into two categories in the following sections of this document: Functional Requirements, which define system behavior, and Non-Functional Requirements, which define system constraints and quality attributes. The system's architecture is then visually mapped using UML diagrams.

FUNCTIONAL REQUIREMENTS:

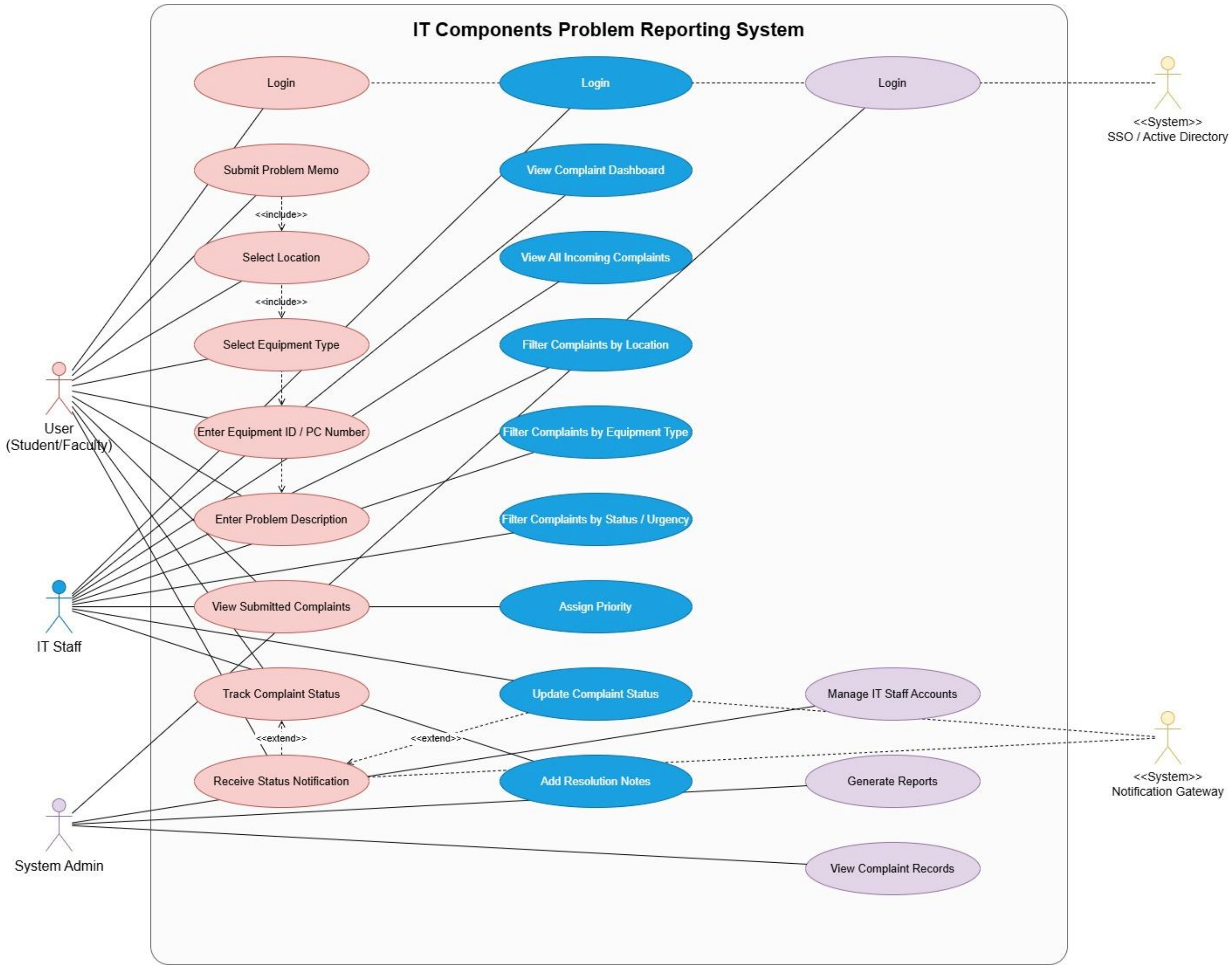
1. The system shall allow users to log in using valid credentials.
2. The system shall allow users to submit a hardware problem memo.
3. The system shall allow users to select the location of the faulty component from a dropdown list.
4. The system shall allow users to select the equipment type from a predefined list.
5. The system shall allow users to enter the equipment ID, PC number, or asset number.
6. The system shall allow users to enter a brief description of the problem.
7. The system shall automatically generate a unique memo ID for each complaint.
8. The system shall store the submitted complaint in a centralized database.
9. The system shall allow users to view all complaints they have submitted.

10. The system shall allow users to track the status of their submitted complaints.
11. The system shall notify users when the status of their complaint changes.
12. The system shall allow IT Staff to view all incoming complaints on a dashboard.
13. The system shall allow IT Staff to filter complaints by location.
14. The system shall allow IT Staff to filter complaints by equipment type.
15. The system shall allow IT Staff to filter complaints by status or urgency.
16. The system shall allow IT Staff to assign a priority level to a complaint.
17. The system shall allow IT Staff to update complaint status as Pending, In Progress, or Resolved.
18. The system shall allow IT Staff to add repair or resolution notes to a complaint.
19. The system shall allow System Admin to generate reports about complaints and resolutions.
20. The system shall allow System Admin to manage IT Staff accounts.

NON-FUNCTIONAL REQUIREMENTS:

1. The system shall provide a simple and user-friendly interface.
2. The system shall be accessible through a web browser.
3. The system shall respond quickly during complaint submission and status checking.
4. The system shall ensure secure login and role-based access.
5. The system shall store complaint data accurately and consistently.
6. The system shall be available during university working hours with minimal downtime.
7. The system shall support multiple users at the same time.
8. The system shall maintain logs of complaint creation and updates.
9. The system shall be easy to maintain and extend in the future.
10. The system shall scale to support more labs, classrooms, and users.

UML DIAGRAM



Use Case1: Login	
Field	Description
Use Case ID	UC-01
Use Case Name	Login
Primary Actors	User (Student/Faculty), IT Staff, System Admin
Description	Allows an authorized actor to log into the system using valid credentials.
Preconditions	The actor must have a registered account in the system.

Postconditions	The actor successfully accesses the system dashboard based on their role.
Main Flow	➢ Actor enters username and password. ➢ System validates credentials. ➢ System identifies actor role. ➢ 4. System redirects actor to the appropriate dashboard.
Alternative Flow	If credentials are incorrect, the system displays an error message and asks the actor to try again.

Use Case 2: Submit Problem Memo	
Field	Description
Use Case ID	UC-02
Use Case Name	Submit Problem Memo
Primary Actor	User (Student/Faculty)
Description	Allows a user to report a hardware problem through a digital memo form.
Preconditions	User must be logged into the system.
Postconditions	Complaint is stored in the system and assigned a unique memo ID.
Main Flow	➢ User opens complaint submission form. ➢ User selects location from dropdown. ➢ User selects equipment type. ➢ User enters equipment ID or PC number. ➢ User writes a short problem description. ➢ User submits the form. ➢ System validates the information. ➢ 8. System stores the complaint and generates a memo ID.
Alternative Flow	If required information is missing, the system prompts the user to complete the form.

Use Case 3: View Submitted Complaints	
Field	Description
Use Case ID	UC-03
Use Case Name	View Submitted Complaints
Primary Actor	User
Description	Allows users to view the list of complaints they have submitted.
Preconditions	User must be logged in and must have previously submitted complaints.
Postconditions	List of complaints is displayed to the user.
Main Flow	➢ User navigates to complaint history. ➢ System retrieves user's complaint records. ➢ 3. System displays the list of submitted complaints.

Use Case 4: Track Complaint Status	
Field	Description
Use Case ID	UC-04
Use Case Name	Track Complaint Status
Primary Actor	User
Description	Allows the user to monitor the progress of a reported complaint.
Preconditions	User must be logged in and complaint must exist.
Postconditions	Current status of the complaint is displayed.
Main Flow	➢ User selects a complaint from their list. ➢ System retrieves complaint details. ➢ 3. System displays the complaint status and updates.

Use Case 5: View Complaint Dashboard	
Field	Description
Use Case ID	UC-05

Use Case Name	View Complaint Dashboard
Primary Actor	IT Staff
Description	Allows IT Staff to see all incoming hardware complaints in a centralized dashboard.
Preconditions	IT Staff must be logged into the system.
Postconditions	All complaints are displayed on the dashboard.
Main Flow	➢ IT Staff logs into the system. ➢ IT Staff opens the complaint dashboard. ➢ 3. System displays all complaints and their statuses.

Use Case 6: Filter Complaints	
Field	Description
Use Case ID	UC-06
Use Case Name	Filter Complaints
Primary Actor	IT Staff
Description	Allows IT Staff to filter complaints based on location, equipment type, or status.
Preconditions	IT Staff must be logged in and complaints must exist.
Postconditions	Filtered complaint results are displayed.
Main Flow	➢ IT Staff selects filter criteria. ➢ System processes filter request. ➢ 3. System displays filtered complaints.

Use Case 7: Assign Priority	
Field	Description
Use Case ID	UC-07
Use Case Name	Assign Priority
Primary Actor	IT Staff
Description	Allows IT Staff to assign urgency level to a complaint.

Preconditions	IT Staff must be logged in and complaint must exist.
Postconditions	Complaint priority level is updated.
Main Flow	➢ IT Staff opens complaint details. ➢ IT Staff selects priority level (Low, Medium, High). ➢ 3. System saves the priority level.

Use Case 8: Update Complaint Status	
Field	Description
Use Case ID	UC-08
Use Case Name	Update Complaint Status
Primary Actor	IT Staff
Description	Allows IT Staff to update the status of a complaint.
Preconditions	IT Staff must be logged in and complaint must exist.
Postconditions	Complaint status is updated and user is notified.
Main Flow	➢ IT Staff selects a complaint. ➢ IT Staff updates the complaint status (Pending, In Progress, Resolved). ➢ System saves the status update. ➢ 4. System sends notification to the user.

Use Case 9: Add Resolution Notes	
Field	Description
Use Case ID	UC-09
Use Case Name	Add Resolution Notes
Primary Actor	IT Staff
Description	Allows IT Staff to add repair or troubleshooting notes to a complaint.
Preconditions	IT Staff must be logged in and complaint must exist.
Postconditions	Resolution notes are stored in the system.

Main Flow	➢ IT Staff opens complaint record. ➢ IT Staff enters resolution notes. ➢ 3. System saves the notes.

Use Case 10: Generate Reports	
Field	Description
Use Case ID	UC-10
Use Case Name	Generate Reports
Primary Actor	System Admin
Description	Allows admin to generate reports on complaint trends and resolutions.
Preconditions	Admin must be logged in and complaint records must exist.
Postconditions	Report is generated and displayed or exported.
Main Flow	➢ Admin opens reporting module. ➢ Admin selects report type and filters. ➢ System gathers complaint data. ➢ 4. System generates the report.

Use Case 11: Manage IT Staff Accounts	
Field	Description
Use Case ID	UC-11
Use Case Name	Manage IT Staff Accounts
Primary Actor	System Admin
Description	Allows admin to create, update, or remove IT Staff accounts.
Preconditions	Admin must be logged in.
Postconditions	IT Staff account information is updated in the system.
Main Flow	➢ Admin opens staff management module. ➢ Admin selects create, update, or delete action. ➢ Admin enters or modifies staff details.

	➢ 4. System saves the changes.